

Thursday, April 3, 2025

Section 6.4

LEC35 - Working With Taylor Series

Warm Up Problem

Which of the following series can be the beginning of the Taylor series of f , centered at $a = 3$?

$\times P(x) = 5.2 + 1.2(x-3) - 0.2(x-3)^2 \dots$ $f(3) \neq 5.2$

$\times Q(x) = 1.7 - 1.1(x-3) - 0.3(x-3)^2 \dots$ $f'(3) \neq 0$

$\times R(x) = 1.6 + 1.2(x-3) + 0.3(x-3)^2 \dots$ $f''(3) \neq 0$

$S(x) = 1.6 + 1.1(x-3) - 0.2(x-3)^2 \dots$

Example

Use three non-zero terms of a Maclaurin series to approximate

$$\int_0^1 \frac{\sin(x)}{x} dx, \text{ where } \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, R = \infty$$

$$\Rightarrow \int_0^1 \frac{\sin x}{x} dx = \int_0^1 \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!} dx = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)(2n+1)!}, \text{ where } t \neq 1$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)(2n+1)!} \approx \frac{(-1)^0}{1 \cdot 1} + \frac{(-1)^1}{3! \cdot 3} + \frac{(-1)^2}{5! \cdot 5} = 1 - \frac{1}{18} + \frac{1}{600} \approx 0.946$$